

WASP-2b

R-0102

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_171024 · created 2026-07-16T03:15:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458050.6114 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.036
Transit depth (Rp/Rs)^2	1.73 +/- 0.95 [%]
Orbital Inclination (inc)	84.04 +/- 1.41
Airmass coefficient 1 (a1)	0.96 +/- 0.011
Airmass coefficient 2 (a2)	0.034 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#1 - [244, 431]
Optimal Aperture	4.1
Optimal Annulus	4.1
Transit Duration (day)	0.054 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.036 vs 0.1326 (deviation 0.04σ)
Duration fit vs published	0.054 d vs 0.0737 d (deviation 0.8σ)
Mid-time O–C	7.7 min
Depth SNR	3.2
Residual scatter	1.13 %

Light curve

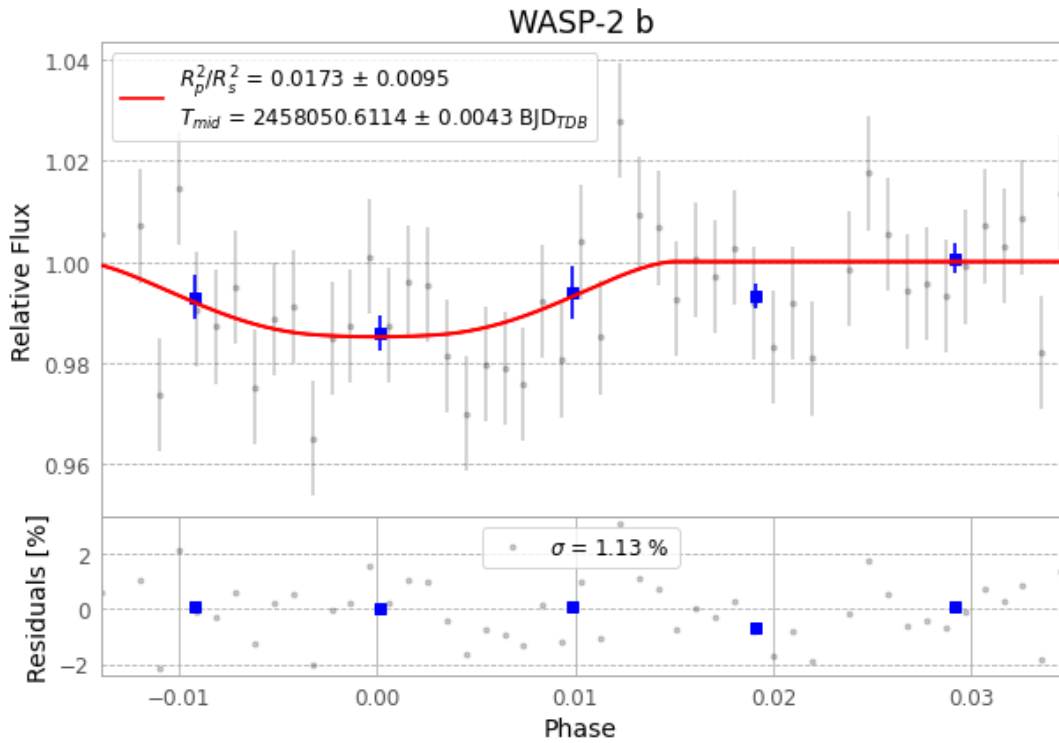


Figure 1 — FinalLightCurve_WASP-2 b_2017-10-23.png (SHA-256 0c77ad2b2dd1ec4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1a0590e5dc5ccbbf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

50 FITS frames (33 MB), WASP-2171024015410.FITS → WASP-2171024042412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-171024.log (ed1f729a6b7a...), FinalLightCurve_WASP-2 b_2017-10-23.png (0c77ad2b2dd1...), FinalLightCurve_WASP-2 b_2017-10-23.csv (3620c80cbd3f...)

Compute resources

Wall time 355.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:15Z.